

Diencephalon I (The THALAMUS)

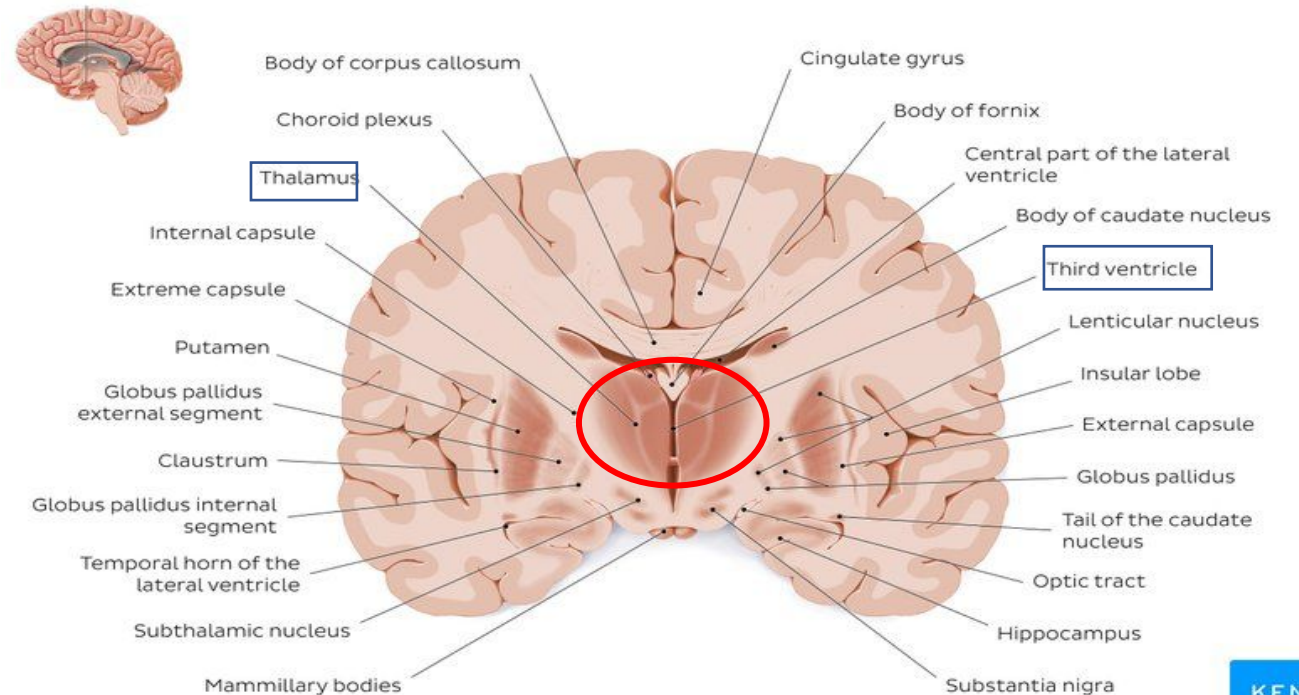
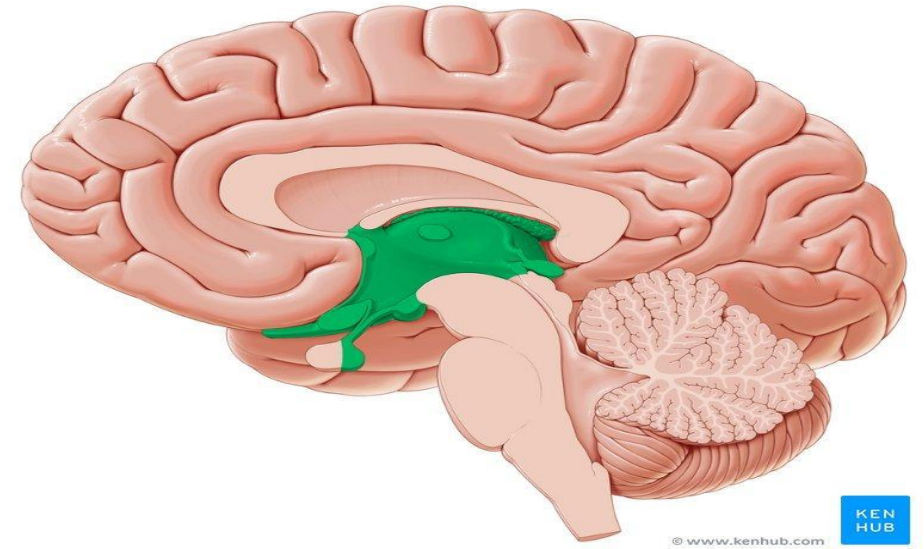


Learning objectives:

- At the end of the lecture, the students should be able to define:
- Diencephalon and its parts/divisions.
- The thalamus, its location and extent.
- Subdivisions of thalamus and names of important nuclei present in each subdivision.
- Ventral posteromedial (VPM) and ventral posterolateral (VPL) nuclei & their connections.
- Medial and lateral geniculate bodies and their associated pathways.

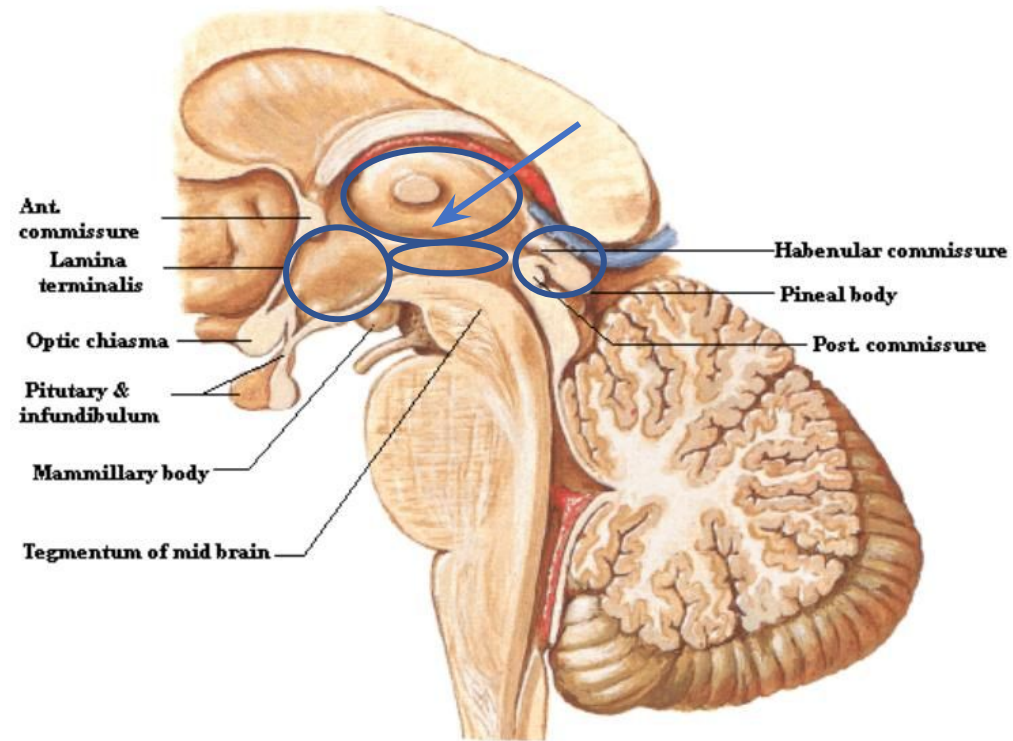
The Diencephalon:

- It consists of third ventricle and the structures forming its boundaries.
- It is a midline structure with symmetrical right & left halves.
- It extends anteriorly to the interventricular foramina and posteriorly to the cerebral aqueduct.
- The diencephalon can be divided into four major parts: (1) the thalamus, (2) the subthalamus, (3) epithalamus and (4) the hypothalamus.



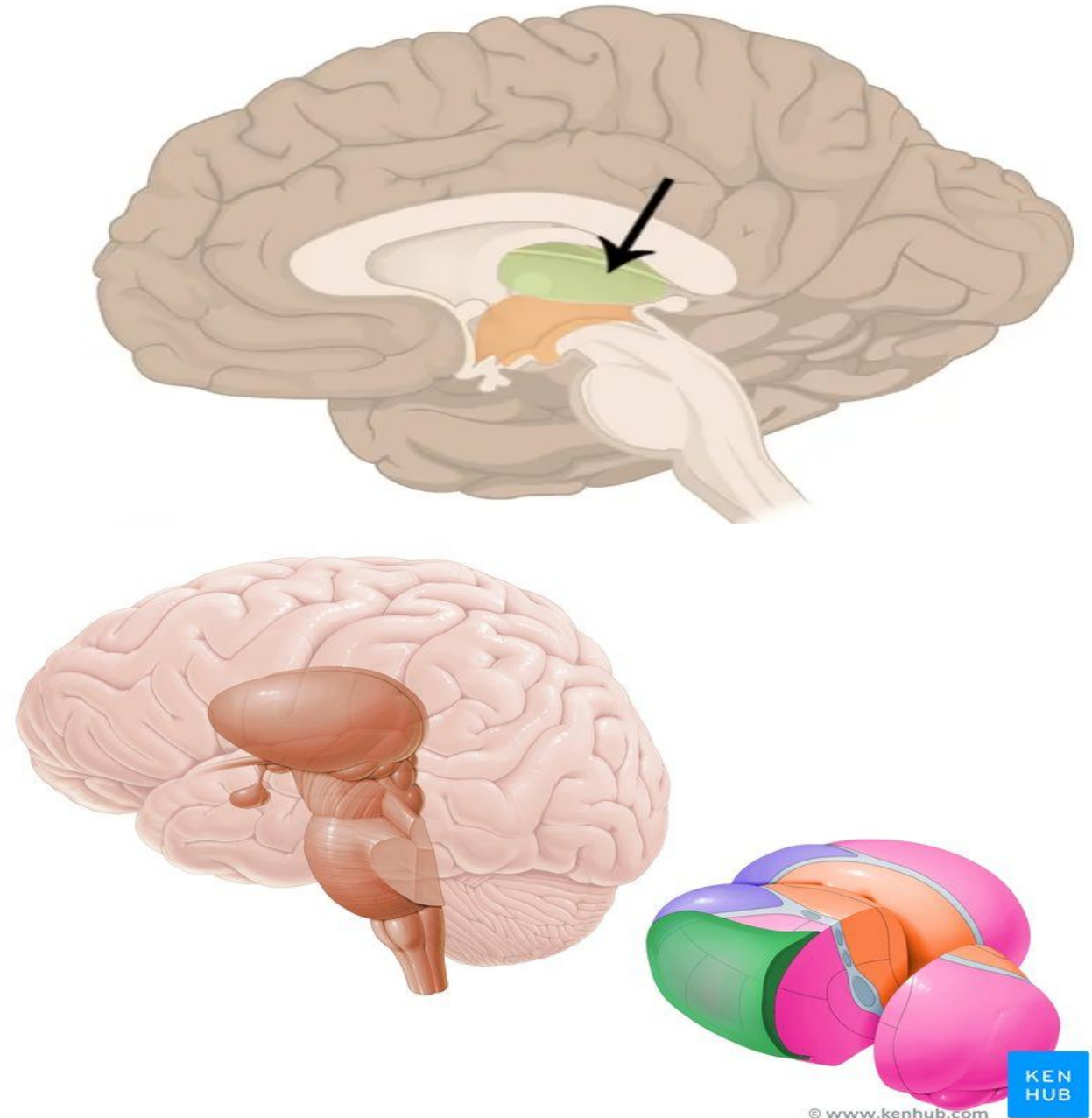
Diencephalon

- From interventricular foramen to posterior commissure
- Divisible in to:
 - Thalamus
 - Hypothalamus
 - Subthalamus
 - Epithalamus
 - Metathalamus

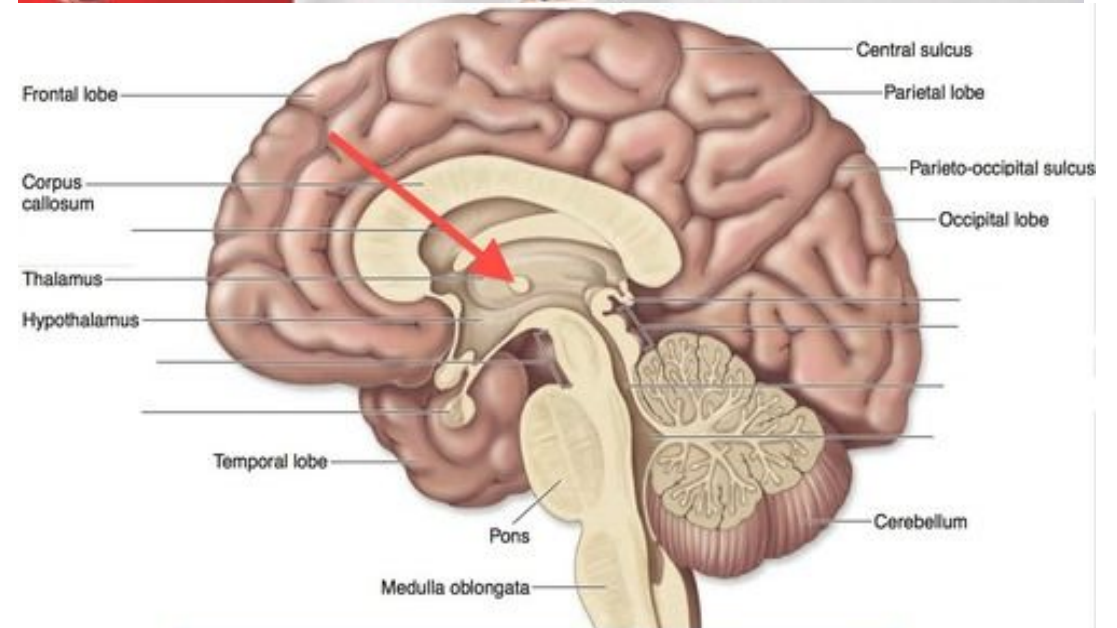
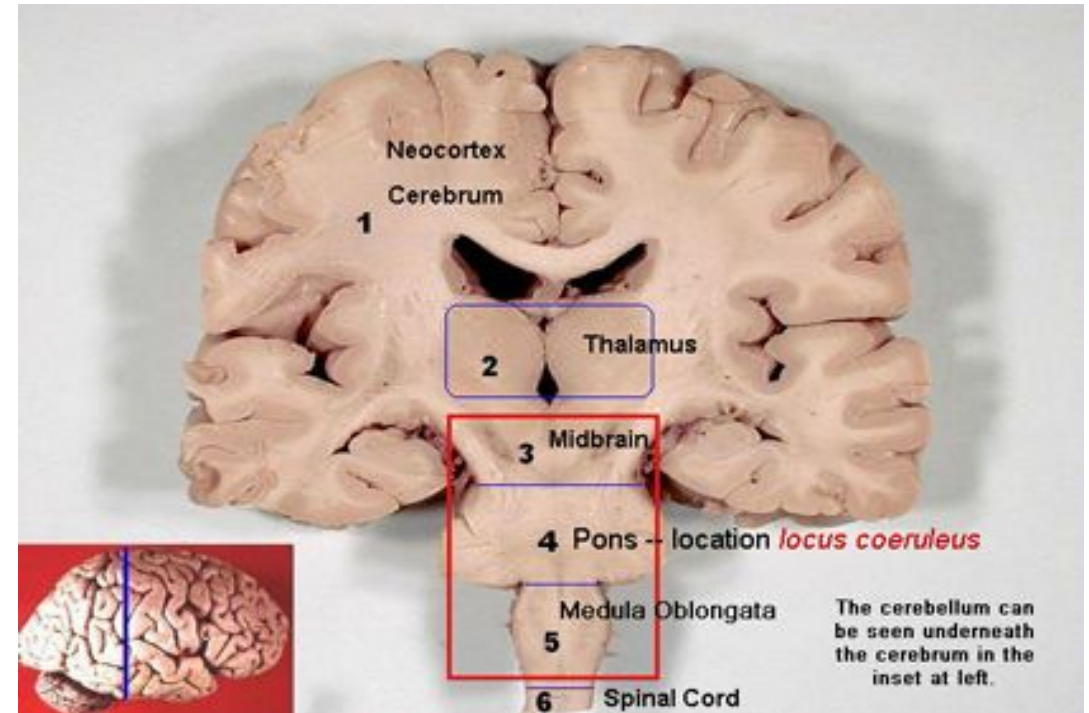


The Thalamus:

- It is a large ovoid mass of gray matter which forms the major bulk of diencephalon.
- It is situated on each side of third ventricle.
- Its anterior end is narrow & related to interventricular foramen.
- Posterior end is expanded to form pulvinar.
- Superior colliculus & superior brachium lie underneath pulvinar.

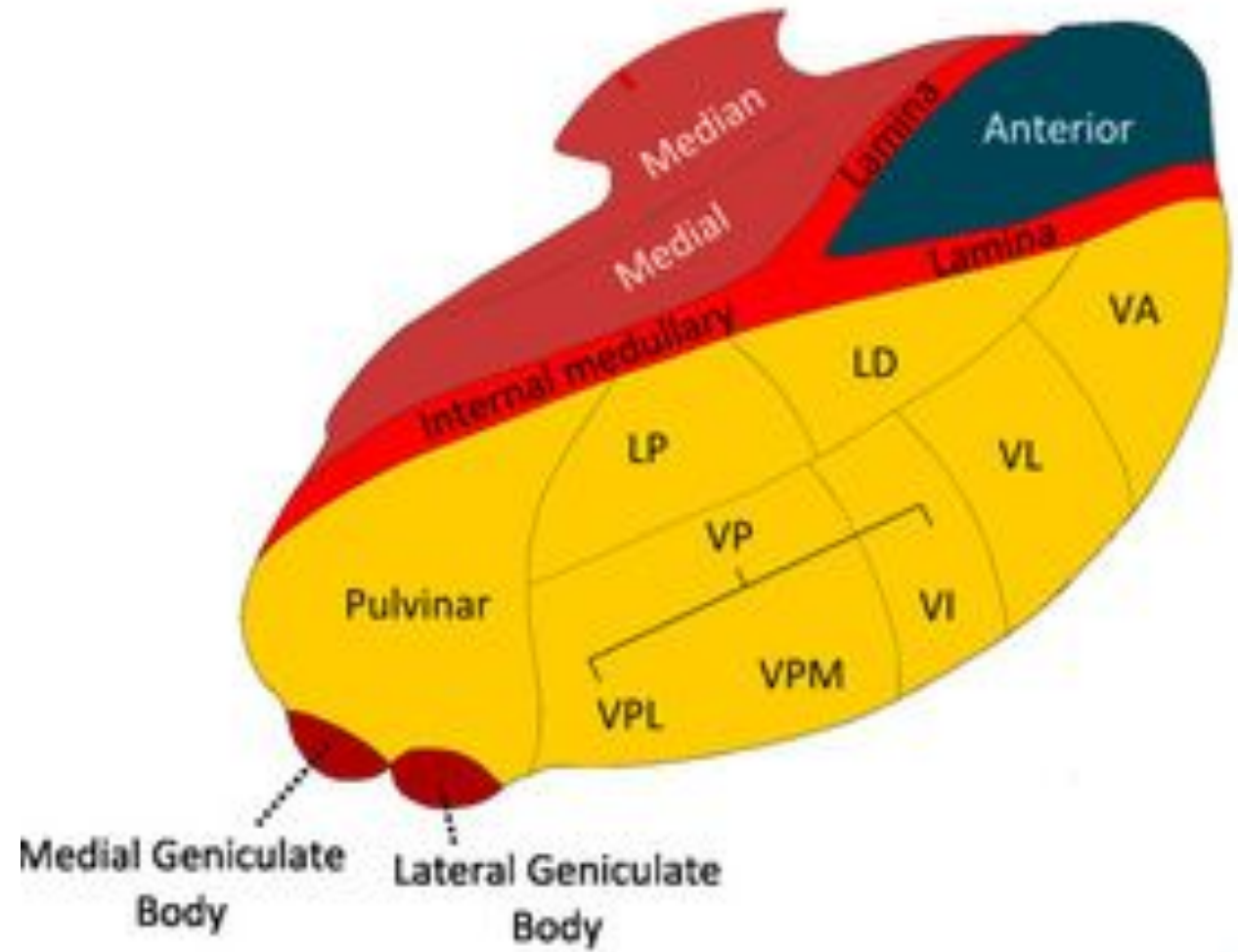


- Superior surface is related to the structures of lateral ventricle.
- Inferior surface is continuous with tegmentum of midbrain.
- Medial surface forms lateral wall of third ventricle and,
- Lateral surface is separated from lentiform nucleus by a band of white matter, the internal capsule.
- Thalamus of one side is connected to that of opposite side by a band of gray matter, the interthalamic adhesion/connection.



Subdivisions of Thalamus:

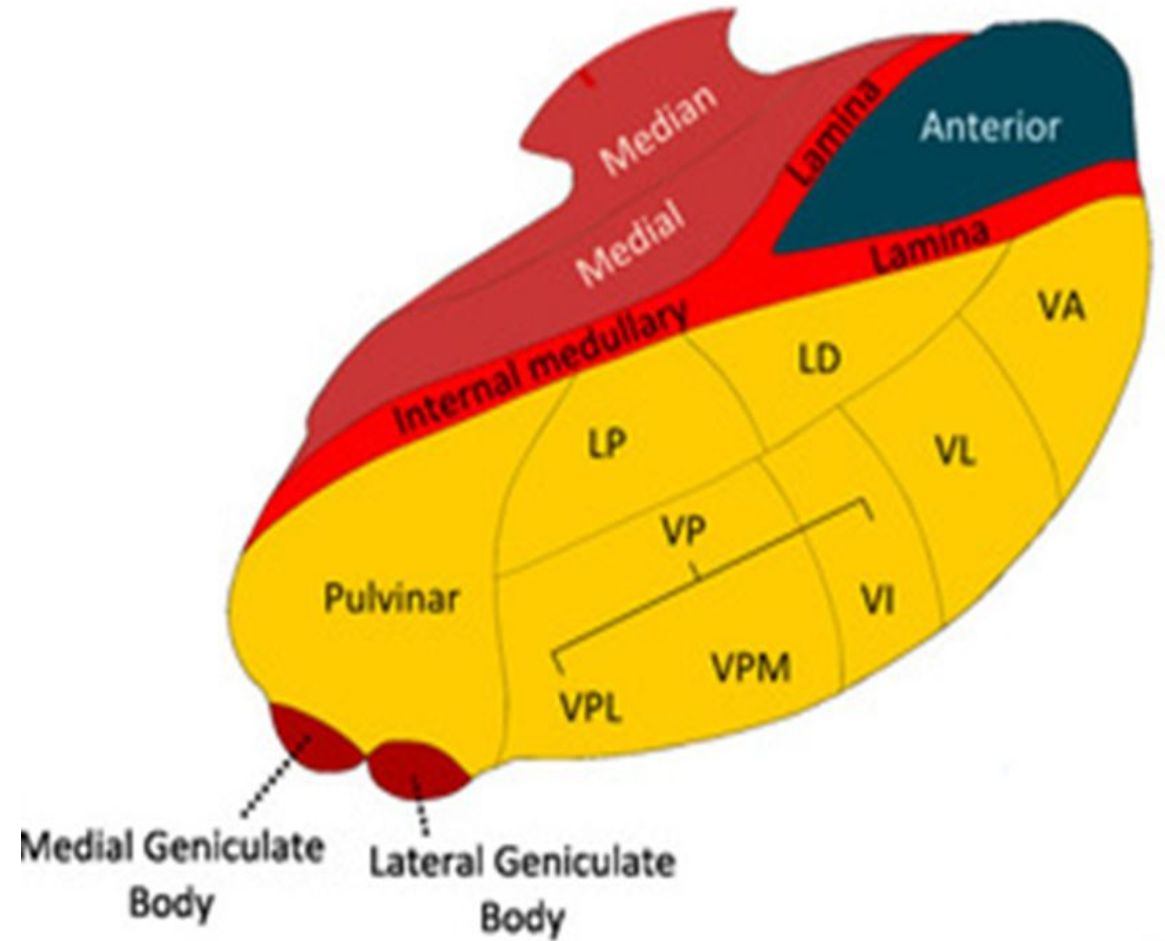
- Superior surface of thalamus is covered by stratum zonale, a layer of white matter.
- Lateral surface is covered by another layer of white matter, the external medullary lamina.
- Gray matter of thalamus is divided into medial and lateral parts by a vertical sheet of white matter, the internal medullary lamina.
- Internal medullary lamina splits and resembles a 'Y'.



- Thus the internal medullary lamina divides the thalamus into anterior, medial & lateral parts.
- Each part contains a group of thalamic nuclei.

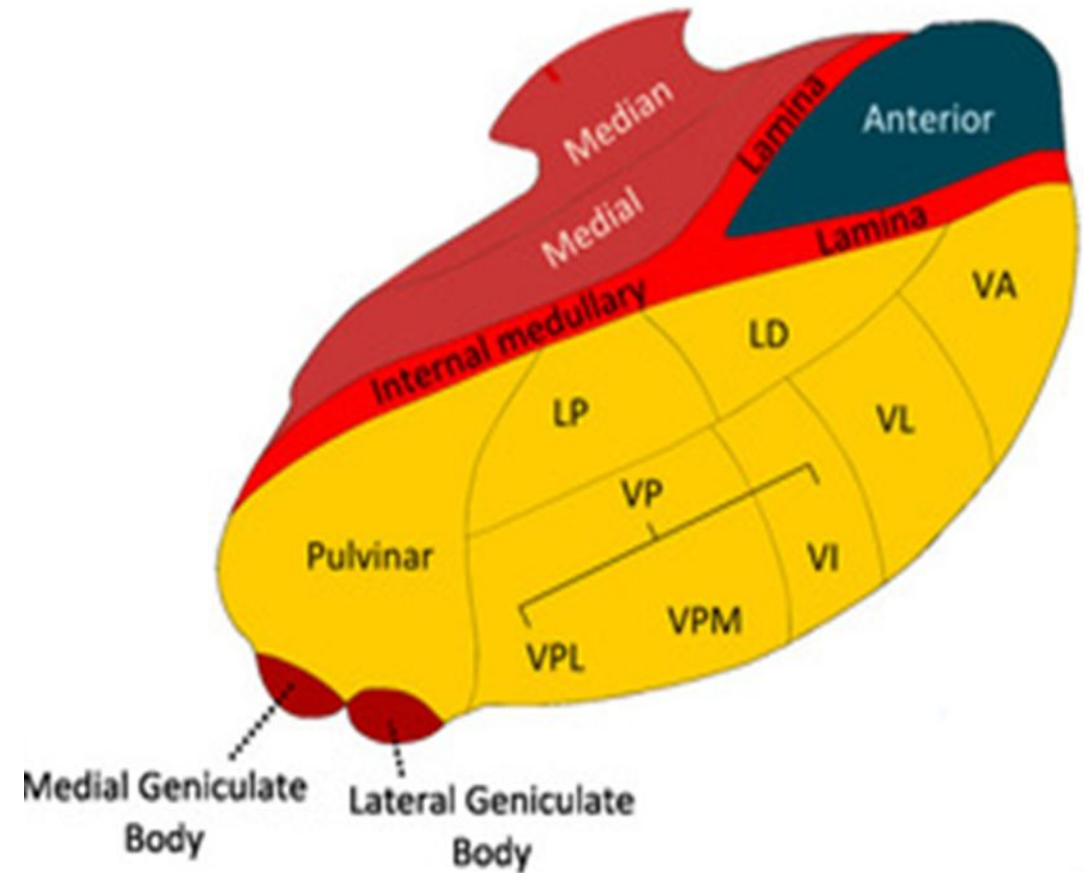
Anterior part:

- Contains anterior thalamic nuclei.
- They receive connections from mamillary nuclei, cingulate gyrus and hypothalamus.
- Concerned with emotional tone and mechanisms of recent memory.



Medial part:

- Consists of a large dorsomedial nucleus and other smaller nuclei.
- The nucleus has connections with whole prefrontal cortex of frontal lobe and hypothalamic nuclei.
- This part of thalamus is responsible for integration of large variety of sensory information and relation of this information to one's emotional feelings.



Lateral part:

- The nuclei are divided into a dorsal tier and a ventral tier.

Dorsal tier of nuclei:

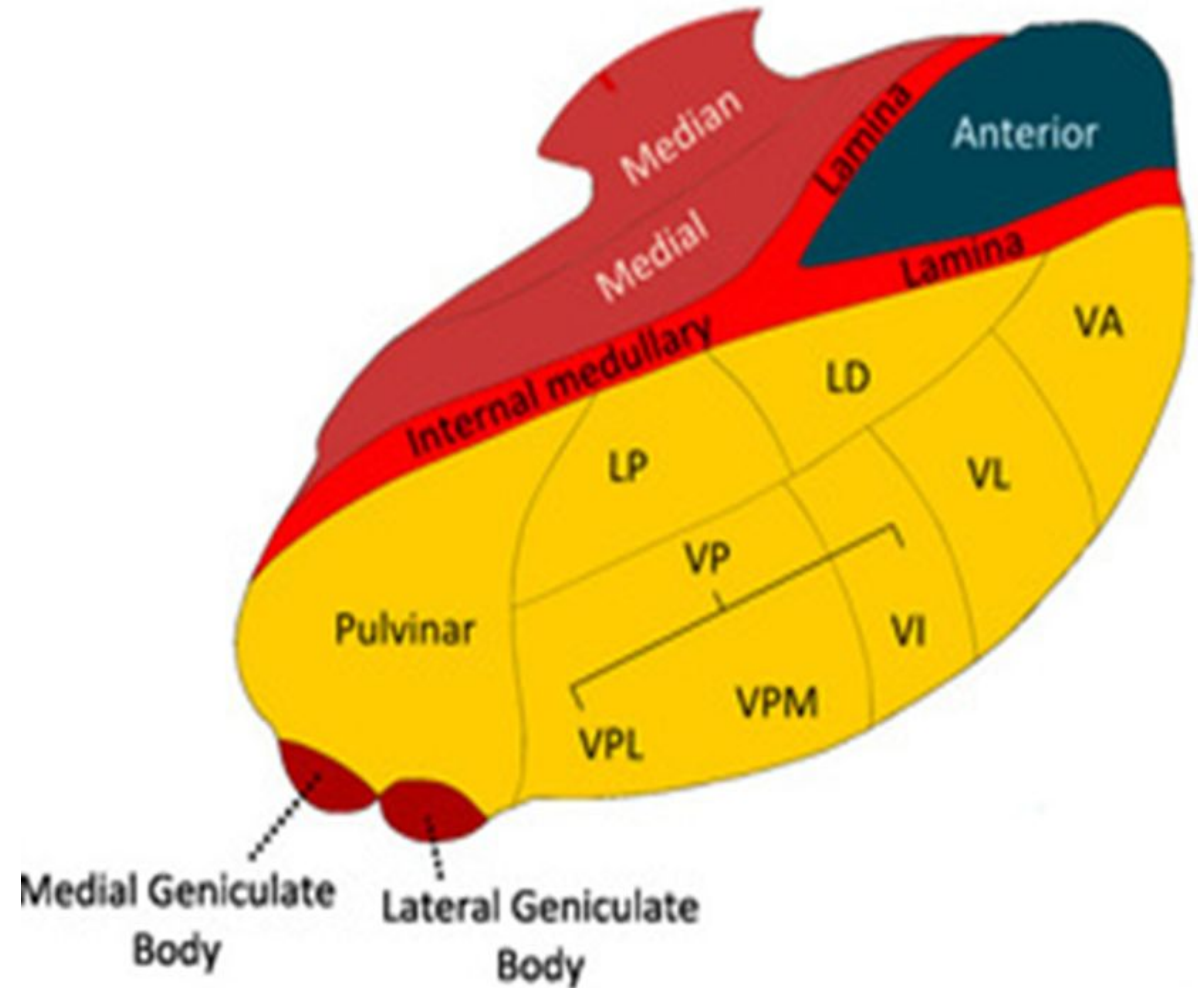
- Contains lateral dorsal (LD) nucleus, lateral posterior (LP) nucleus and pulvinar.

Ventral tier of nuclei:

- Consists of the following nuclei in craniocaudal sequence:

I. Ventral anterior (VA) nucleus:

- It has connections with reticular formation, substantia nigra, corpus striatum & premotor cortex.
- It lies on the pathway between corpus striatum and motor areas of frontal cortex and influences the activities of motor cortex.

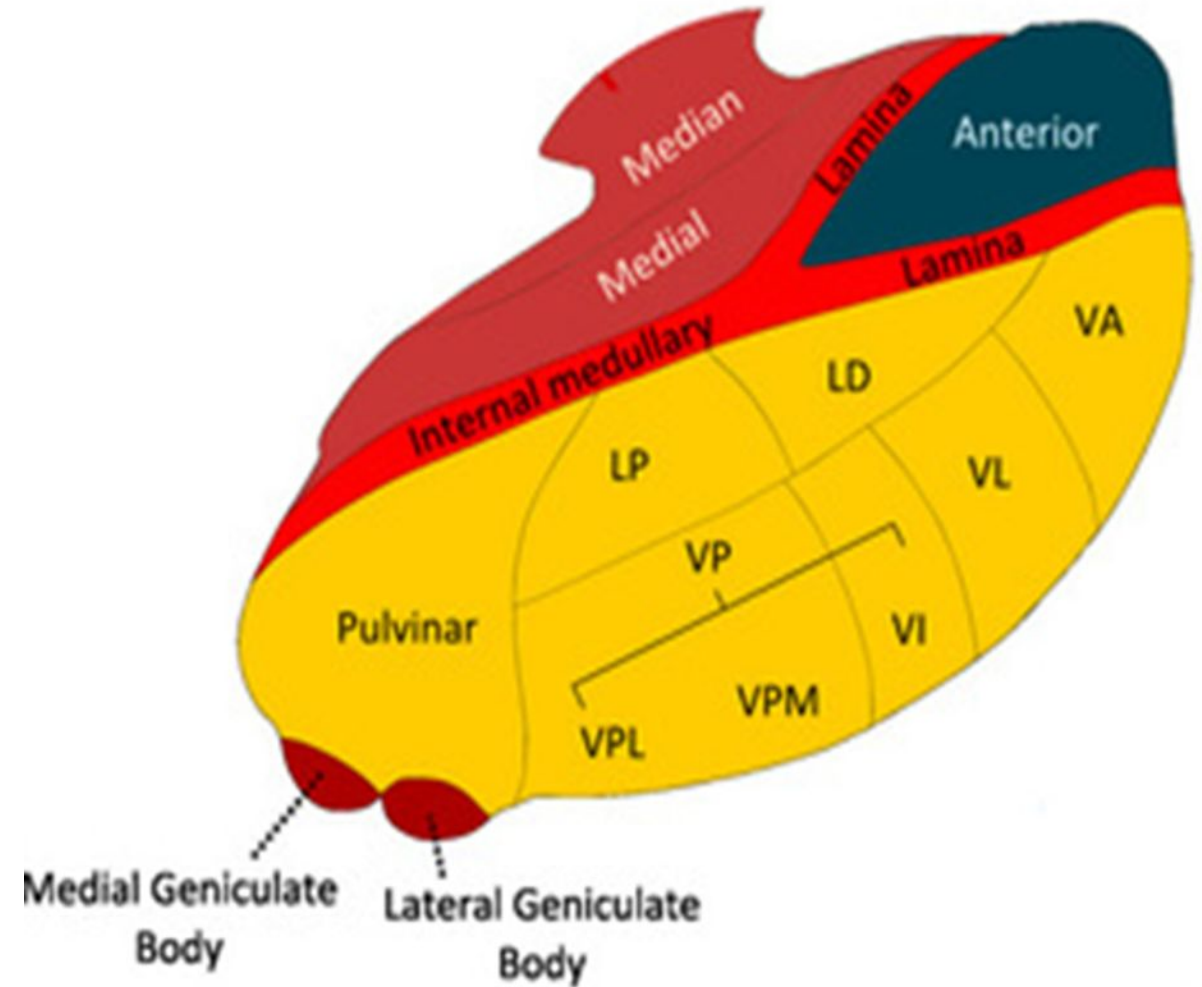


II. Ventral lateral (VL) nucleus:

- Receives major input from cerebellum and minor input from red nucleus.
- Projections pass to motor and premotor areas of cerebral cortex.
- Influences motor activity.

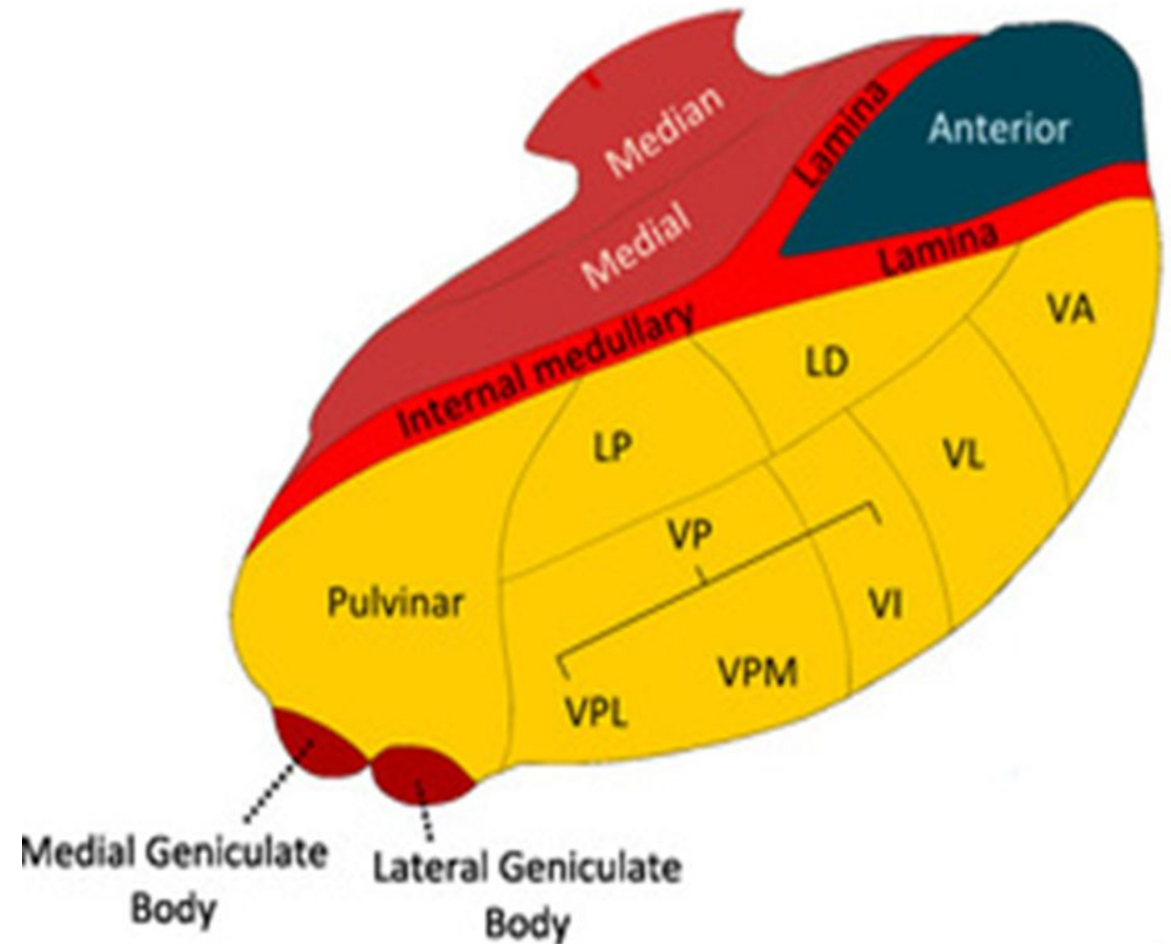
III. Ventral posterior (VP) nucleus:

- Subdivided into ventral posteromedial (**VPM**) nucleus and ventral posterolateral (**VPL**) nucleus.



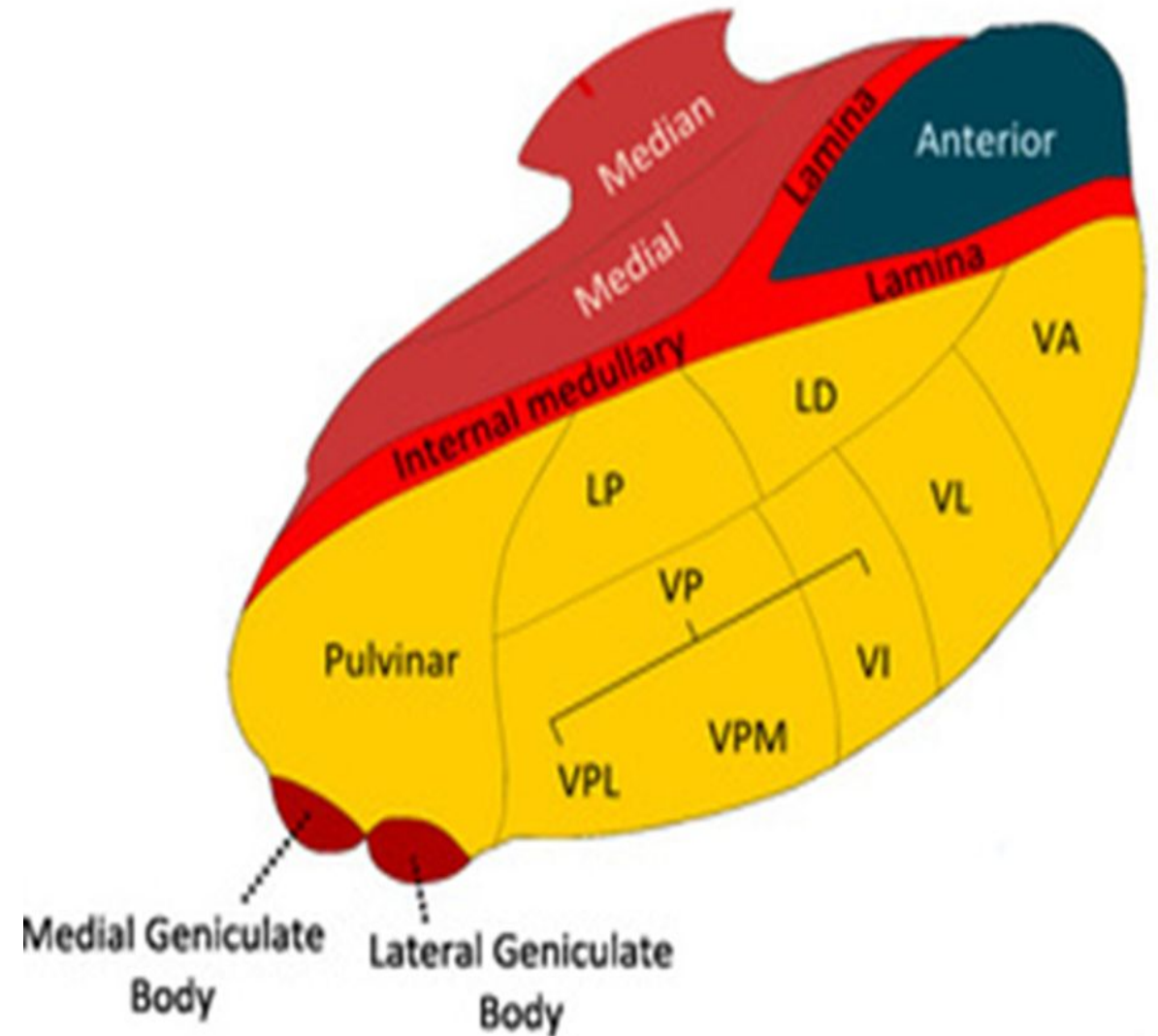
III. Ventral posterior (VP) nucleus: (continued)

- VPM nucleus receives ascending trigeminal and gustatory pathways.
- Whereas VPL nucleus receives important ascending tracts, the medial and spinal lemnisci.
- The thalamocortical projections pass through corona radiata & internal capsule to somatosensory cortex (postcentral gyrus, areas 3, 1, & 2).



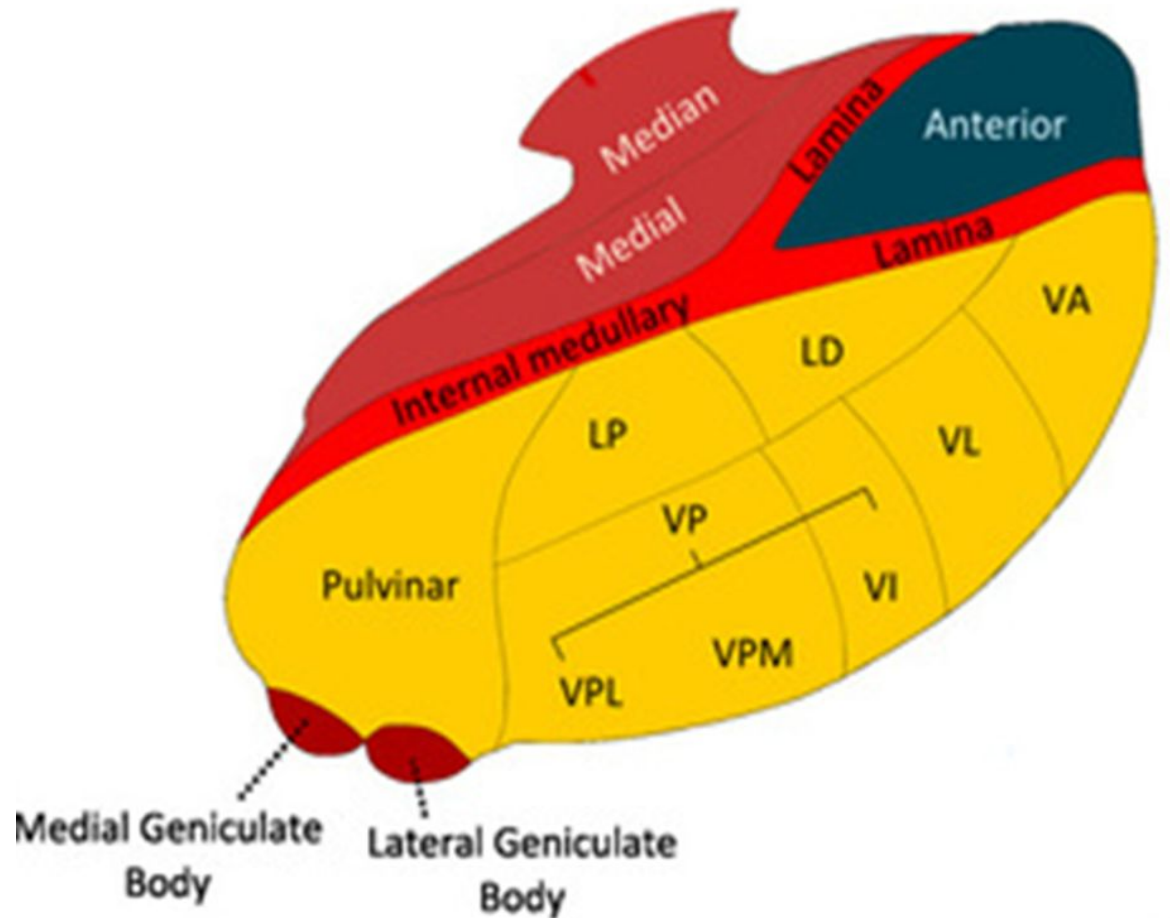
Other thalamic nuclei:

- I. **The intralaminar nuclei:** consist of small collection of nerve cells within internal medullary lamina.
- II. **Midline nuclei:** located adjacent to third ventricle and interthalamic adhesion.
- III. **Reticular nucleus:** located between external medullary lamina & internal capsule (posterior limb).



IV. Medial geniculate body (part of auditory pathway):

- Receives afferent fibers from inferior colliculus through inferior brachium.
- Inferior colliculus receives termination of fibers of lateral lemniscus.
- Medial geniculate body receives auditory information from both ears but predominantly from opposite ear.
- Efferent fibers form auditory radiation which reaches the auditory cortex (superior temporal gyrus).



V. Lateral geniculate body (part of visual pathway):

- Present as a swelling on the undersurface of pulvinar.
- Almost all the fibers of optic tract terminate here except few passing to pretectal nucleus.
- Afferent fibers come from ganglion cell layer of retina from temporal half of same side's eye & nasal half of opposite side's eye, thus latter fibers cross midline in optic chiasma.
- Efferent fibers form visual radiation which reaches the visual cortex of occipital lobe.

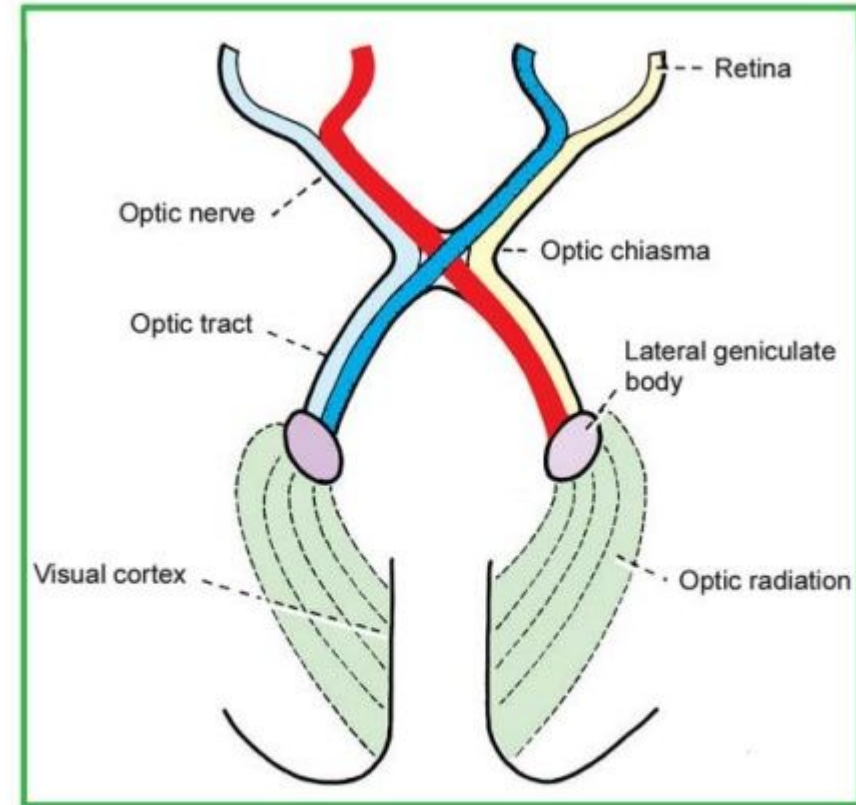


Fig. 18.1. The optic pathway. Note that the fibres from the medial (or nasal) half of each retina cross over to the optic tract of the opposite side.

Functions of Thalamus

- Sensory integration and relay station for all the sensory pathways except Olfactory
- Recognition of crude pain, temperature & touch
- Influences voluntary movements by receiving impulses from basal ganglia & cerebellum & relaying them to motor cortex , influence LMN
- Participates in maintenance of state of wakefulness & alertness through RAS
- Role in emotions & recent memory

Important Thalamic Nuclei, Their Nervous Connections and Their Functions			
Thalamic Nucleus	Afferent neuronal loop	Efferent neuronal loop	Function
Ventral Posteromedial (VPM)	Trigeminal lemniscus, gustatory fibers	Primary somatic sensory cortex (areas 3,1, & 2)	Relays common sensations to consciousness
Ventral Posterolateral (VPL)	Medial and spinal lemnisci	Primary somatic sensory cortex (areas 3,1, & 2)	Relays common sensations to consciousness
Medial geniculate body	Inferior colliculus, lateral lemniscus from both ears but predominantly the contralateral ear	Auditory radiation to superior temporal gyrus	Hearing
Lateral geniculate body	Optic tract	Optic radiation to visual cortex of occipital lobe	Visual information from opposite field of vision

References:

CLINICAL NEUROANATOMY

Richard S. Snell

8th edition

Chapter 12

Pages # 372-376